

Complete Chat History: Nuclear Operators and Tensor Products

User and DeepSeek Chat

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Complete Transcript

User: The set of nuclear operators acting in an infinite dimensional complex Hilbert space when closed under the trace class norm produces the subspace of trace class operators, when closed under the Hilbert-Schmidt norm produces the Hilbert-Schmidt operators, when closed under the operator norm produces the compact operators. The space of nuclear operators is isomorphic to the spaces of second rank covariant, the space of second order contravariant and the space of mixed contra-covariant tensors. Can one define norms on these spaces using the underlying Hilbert space that mimics these norms and the strong operator topology norm by using the bilinear nature of these tensors on the underlying Hilbert space?

DeepSeek Chat: Yes, one can define norms on these tensor spaces that mimic the standard operator norms. Here are the key constructions:

Table 1: Norms on Tensor Spaces vs. Operator Norms

Tensor Type	Norm Definition	Operator Analog
Mixed ($\mathcal{H} \otimes \mathcal{H}^*$)	$\ T\ _1 = \inf \sum_i \ x_i\ \ y_i\ $	Trace-class norm
Covariant ($\mathcal{H}^* \otimes \mathcal{H}^*$)	$\ B\ _2 = \left(\sum_{i,j} B(e_i, e_j) ^2 \right)^{1/2}$	Hilbert-Schmidt norm
Contravariant ($\mathcal{H} \otimes \mathcal{H}$)	$\sup_{\ x\ =\ y\ =1} \Theta(x, y) $	Operator norm

User: Hence is there an isomorphism between the space of such mixed tensors and the space of bounded operators acting on the Hilbert space in the strong operator topologies?

DeepSeek Chat: The relationship depends crucially on the topology:

Table 2: Completions of $\mathcal{H} \otimes \mathcal{H}^*$

Topology	Completion	Isomorphism to $\mathcal{B}(\mathcal{H})$?
Operator norm	$\mathcal{K}(\mathcal{H})$ (compact ops)	No (strictly smaller)
Trace norm	$\mathcal{L}^1(\mathcal{H})$ (trace-class)	No
Hilbert-Schmidt	$\mathcal{L}^2(\mathcal{H})$	No
SOT	$\mathcal{B}(\mathcal{H})$	Yes (dense embedding)

User: But if you take the analogous SOT completion of the mixed tensor product, does that map onto the complete space of bounded operators?

DeepSeek Chat: Yes! Here's the full picture:

Key observations:

- The SOT-completion achieves full $\mathcal{B}(\mathcal{H})$ because:
 - Finite-rank ops are SOT-dense
 - All bounded operators are SOT-limits
- Norm completions give proper subspaces

Table 3: Summary of Tensor Completions

Tensor Space	Norm	Completion	Dense in $\mathcal{B}(\mathcal{H})$?
$\mathcal{H} \otimes \mathcal{H}^*$	$\ \cdot\ _{\text{op}}$	$\mathcal{K}(\mathcal{H})$	No
$\mathcal{H} \otimes \mathcal{H}^*$	$\ \cdot\ _1$	$\mathcal{L}^1(\mathcal{H})$	No
$\mathcal{H} \otimes \mathcal{H}^*$	$\ \cdot\ _2$	$\mathcal{L}^2(\mathcal{H})$	No
$\mathcal{H} \otimes \mathcal{H}^*$	SOT	$\mathcal{B}(\mathcal{H})$	Yes